Clinical Oncology 29 (2017) 557–561

Contents lists available at ScienceDirect

Clinical Oncology

journal homepage: www.clinicaloncologyonline.net

Overview

Lessons Learnt from Past Incidents and Accidents in Radiation Oncology

T. Knöös [*] [†]

[*] Department of Haematology, Oncology and Radiation Physics, Skåne University Hospital, Lund, Sweden

[†] Department of Medical Radiation Physics, Clinical Sciences, Lund University, Lund, Sweden

Received 2 May 2017; received in revised form 17 May 2017; accepted 30 May 2017

Abstract

The purpose of this report is to review and compile what have been and can be learnt from incidents and accidents in radiation oncology, especially in external beam and brachytherapy. Some major accidents from the last 20 years will be discussed. The relationship between major events and minor or so-called near misses is mentioned, leading to the next topic of exploring the knowledge hidden among them. The main lessons learnt from the discussion here and elsewhere are that a well-functioning and safe radiotherapy department should help staff to work with awareness and alertness and that documentation and procedures should be in place and known by everyone. It also requires that trained and educated staff with the required competences are in place and, finally, functions and responsibilities are defined and well known.

Key words: Accidents; incidents; lessons; radiation oncology; radiotherapy; safety

Introduction

Radiotherapy is a high-tech treatment modality for a large proportion of all cancer patients. It is known that for about 50% of all patients, radiotherapy would be beneficial, either as a primary treatment or palliative (see for example [1]). In 2012, there were about 14 million new cancer cases in the world [2] and if radiotherapy were available for all patients requiring it, we would have treated around seven million patients. Unfortunately, radiotherapy is not available for everybody in the world. Today, radiotherapy is given at around 13 000 treatment units (1900 of these are ^{60}Co units) with a total capacity of 115 million fractions annually [3]. Even if this number is not reached in reality, one may summarise that a large number of radiotherapy fractions are administrated every day, with many opportunities for the complex process to go wrong. Many near misses are present and corrected online by competent staff, but, as can be seen in this review, sometimes the holes in Reason's cheese model align and severe events happens [4].

Review of Some Accidents

Calibration

Exeter, UK

After replacing a ^{60}Co source, the determination of output (dose rate in Gy/min) is among several things that has to be carried out to assure future clinical usage of the treatment machine. In 1998, at Royal Devon and Exeter Hospital, UK, a new source was installed and the dose rate determination was carried out by a medical physicist who unfortunately entered the wrong measurement time in his calculations, resulting in a dose rate 25% higher than planned being used and leading to patient overdose. During the months until this deviation was detected, 153 patients were treated [5].

Due to local circumstances, this was not clinically detected, but instead during a national dosimetry audit (organised by the Institute of Physical Sciences in Medicine), this deviation was detected [6–8]. Notable is that

Author for correspondence: Department of Haematology, Oncology and Radiation Physics, Skåne University Hospital, SE-221 85 Lund, Sweden.

E-mail address: tommy.knoos@med.lu.se.

http://dx.doi.org/10.1016/j.clon.2017.06.008

0936-6555/

there is no mention of any independent or secondary check of the dose rate. Everything depends on a single physicist's actions.

Ottawa, Canada

In 2004, after re-localisation, an orthovoltage machine went through a new commissioning process and a medical physicist carried out the measurements and prepared the data needed for clinical use of the machine. The results for the reference field (i.e. for the cone of 10×10 cm^2) were accurate for all beam qualities available, but when preparing the data for all other cones, a mistake was made and the so-called backscatter factor was left out. This meant that all treatments apart from those with the reference cone were erroneous (in principle by the ratios of backscatter factors between the used cone and the reference).

This error was detected in 2007, when another physicist carried out the annual review and noted some discrepancies in the data. This was followed by extensive measurements and it was revealed that the management of the previous measurements for creating the data for treatment was incorrect. During these 3 years, 620 patients were treated (1019 treatments); 326 were considered to have received a potentially clinically significant underdosage (up to 17%) [9]. Also in this case, there was no documentation regarding any secondary check before the machine was used clinically.

Summary

When errors involving equipment are present in the radiotherapy process, e.g. calibration of the beam, as discussed above, it will involve all patients treated with that unit in a systematic way. In the first case, there was an offset in dose resulting in a 25% overdose to all patients; in the other case, depending on the prescribed field size, the deviation was not equal for all.

Many contributing factors can be identified and discussed in both accidents. However, one lesson to learn is that a new or re-commissioned treatment unit must be checked and rechecked. Unfortunately, this problem is not isolated to only these two events. This has happened on many other occasions [10–12] and will continue to happen. Thus, having an in-house or preferably an external audit of dosimetry before clinical use may catch these problems and consequently improve safety substantially [13]. Having an *in vivo* dosimetry system that is independently calibrated may also catch a systematic deviation [14].

Procedures, Knowledge and Training

Glasgow, UK

In 2005, a new or updated version of the record and verify system and the treatment planning system (TPS) was introduced and taken into clinical use. The update resulted in the two software systems sharing the same database and omitting manual data transfer as previously carried out. This new environment required the prescription dose to be entered at the start of the planning process and is then followed through the whole process until the patient is brought up on the treatment console. This was new at the hospital, as previously all treatment plans had been created with a prescription dose per fraction of 1 Gy and consequently the settings for treatment were the number of monitor units (MU) to give 1 Gy to the target volume. This was before the upgrade adjusted the treatment console by scaling (multiplying) the 'MU per 1 Gy' with the prescribed dose to receive the correct number of MUs for the prescribed dose.

All procedures were updated such that the correct prescription dose had to be entered when planning started and the scaling of MUs at the treatment unit was no longer necessary. However, one protocol was overlooked – for cranio-spinal treatments. A form remained in use that contained a part where the number of MUs/Gy had to be entered and then rescaled by the prescription dose to receive the final number.

A patient (Lisa Norris) was planned at the end of 2005 and the prescription dose was entered into the TPS as the new routine. At the end, after the plan was approved, the form for cranio-spinal treatments was completed and the final MU to be given to the patient was scaled by the prescription dose of 1.67 Gy, resulting in an overdose of 67%, as it was applied twice.

This patient was only the third patient of this group (out of about 9000 patients annually) since the upgrade of the system. The first had been handled correctly; the second had a prescription dose of 1 Gy/fraction. Another case was planned in January 2006 and was entered and managed correctly. In February a fifth case was planned and it happened to be the same planner as for this case, who then realised the mistake she/he had made filling out the form. At this moment, the patient had just received her 19th fraction and the treatment was interrupted. This description is based on the full report by the Scottish Minister [15].

This case is a consequence of many contributing factors; for example, an inexperienced planner working under supervision, the plan being checked by the planner's supervisor (rather than an independent person), an under-staffed department, written procedures not updated, etc. [16].

Many aspect of what went wrong and what quality controls (or barriers) failed can be discussed, but changing the process requires special attention to all processes and also that all staff are updated and made aware of the changes introduced. For example, why did no-one question the scaling of MU for this particular patient as that procedure was no longer in use?

New York, NY, USA

A patient with head and neck cancer started treatment (intensity-modulated radiotherapy; IMRT) at St Vincent's Hospital in New York City in early 2005 (8 March). The treatment plan had been reviewed and passed the quality controls according to the procedures at the hospital. After the fourth treatment (all four sessions had been delivered without any problem), the responsible radiation oncologist wanted to modify the treatment to reduce the dose to the teeth in the anterior part of the mouth. This was on Friday 11 March. On the following Monday, the physics/dosimetry room was informed about this and started to re-plan the case after the changes initiated by the radiation oncologist.

This led to new IMRT optimisation with changed fractionation, total dose etc. In the software used this is a two-step process: first the required fluence is calculated to fulfil the dose/volume criteria set-up by the planner; second, a process starts where the movements and positions of the field forming device (the multileaf collimator, MLC) is determined. When planning is completed (an accepted dose distribution), all have to be stored to the database. This includes all parameters for treatment, the actual fluence calculated, the generated digital reconstructed radiographs and MLC control points (where each leaf should be positioned and for the number of MUs). During this process, the planning workstation hung (which they frequently do) and the only way to get further was to reboot the computer. To save time, as the patient and family members were in the waiting area, the planner brought up the patient on another work station. The plan was reviewed and approved and the patient was called to the treatment unit where the patient file was opened with the new plan. The patient was treated with this new plan. He mentioned that the treatment did not not feel the same and he had nausea. After the third fraction, on 16 March, the physicist carried out patient-specific quality control according to the local procedures. This was repeated several times (three times) as the results were startling; the fraction dose measured was 13 Gy instead of the prescribed 2 Gy. What the physicist saw was horrifying; the MLC, which was supposed to move to precisely deliver the wanted IMRT to his tumour, was wide open. Consequently, the patient was informed about this dreadful situation and the remaining treatments were cancelled. The patient lived for about 2 years with severe side-effects until he died in February 2007 [11,17].

This extraordinary accident included many different contributing factors leading to the mistreatment of the patient. From the lack of experience and knowledge that IMRT is not 3D conformal radiotherapy (3DCRT) where one can easily change field size by just adjusting MLCs or jaws to not carrying out patient-specific quality control for the new IMRT plan before treatment, i.e. following procedures. Neither was the plan double checked by a second physicist or checked by measurements before treatment, as stated in their quality system. Over-riding barriers in a process may lead to adverse problems. Awareness is also a question here, as during treatment the staff had the chance to notice that the MLC was neither present nor moving on the computer screens for 3 consecutive days. This may also be a question of knowledge and competence. Were the staff at the treatment unit familiar with IMRT treatments?

North Staffordshire, UK

Until 1982, a hospital relied on manual calculations for the correct dose to be delivered to the tumour. Treatments were generally carried out at a standard source–skin distance. Very few treatments were given with a fixed source–axis distance, or as it usually called, isocentric treatment, which today is the standard set-up of the patient. For those cases, the calculation of the exposure time involves a correction for the shorter distance, i.e. the inverse square law has to be applied, reducing the time (or number of MU). A computerised TPS was acquired in 1981 and went into clinical use in the autumn of 1982. Partly because the TPS simplified the calculation procedures, the hospital began treating more and more patients with an isocentric set-up. The old procedure applying the inverse square law for isocentric treatments was still used by the treatment staff before entering the treatment time (minutes or MUs).

Ten years later, in 1991, a new TPS was installed and during commissioning a discrepancy was discovered between the new plans and those from the older system. Further investigation revealed that the previous TPS already carried out an inverse square correction for treatment with other distances than source–skin distance equal to the reference of 100 cm. Thus, in practice, the correction was applied twice, as it was repeated at the treatment unit. During this time interval (9 years), 6% of all patients treated at the department were treated with isocentric technique. As the correction was applied twice, all these patients received too low a dose. For many of these patients, however, it was only part of their treatment. In total, 1045 patients were affected by this procedure, of which 492 developed local recurrences that could be attributed to the error. The underdosage varied between 5 and 35% [18,19].

Summary

For these three events one may see a pattern of unawareness and perhaps also a knowledge/competence problem. A lack of competent staff may also have contributed. Procedures were not updated when radiotherapy systems were replaced and had new features, perhaps unknown to local users because of insufficient training. In the New York case, procedures were in place but were not followed, which may be an indication of a lack of understanding.

Incident Reporting

In the previous part we discussed a few very serious events that may have had a fatal ending. We can also find several other published adverse events, but they are only the tip of the iceberg. This is also illustrated by the Heinrich triangle, which illustrated the ratio between major events to minor events to near misses as 1:30:300 [20]. In the safety report from the International Atomic Energy Agency [12], about 90 events are summarised, from external, brachytherapy and isotope therapy. However, what still has to be realised is that very few events occur at an individual hospital. Thus, there is a need to collect and analyse minor deviations (incidents and near incidents/misses) in the radiotherapy process and having systems where this information can be shared among professionals and to learn from these incidents in the context of improved departmental infrastructure and procedures [21].

Conclusions

A theme seen in all accidents reported is that working with awareness and alertness can improve safety and perhaps have avoided the adverse event. Special attention is needed when unusual and complex treatments are carried out. This includes all steps in the process. Is the set-up (e.g. immobilisation) sufficient and will it facilitate the treatment to be given? One should always be alert if unusual circumstances exist during treatment, e.g. patients commenting regarding what is going to be treated (doctor said right side but this is the left!), new or lack of sufficient competent staff for the specific treatment. Another warning signal can be that some treatment parameters are unusual, e.g. many MUs, beam directions.

It has also been noted that comprehensive documentation and procedures must be in place covering acceptance, commissioning, quality control and the daily clinical usage of equipment. Procedures based on national and international recommendations should be in place and for the daily work these have to be tailored to the local conditions. Before equipment is clinically used, independent secondary checks should be carried out. Relying on a single medical physicist's work has several times been shown to be hazardous. Audits carried out internally, but preferably externally, have the potential of improving safety. In this context one should not only think about the dosimetric part; clinical procedures should also regularly be audited.

Following directly from above, having well-trained and educated staff is compulsory to maintain safety in the complex radiotherapy environment. It is necessary to have staff that fully understand the capabilities of today's complex treatment machines and able to take advantage of them. Taking part in the vendor's user training of new equipment (e.g. linacs, TPS, treatment modalities) for staff is the first step to get familiar with the equipment; a second step is participating in professional development using new technologies, e.g. ESTRO school courses and other professional organisations' education programmes. In this context, one can notice the first commandment by Schreiner regarding dosimetry verification of today's complex delivery techniques: 'know and understand your dosimetry system completely, including its limitations, before applying it to a particular validation task' [22].

It is also important to participate in meetings in the field – the second commandment: 'engage in the clinical exchange of ideas and knowledge through publication in scientific journals, and, perhaps more importantly, through regular communication, meetings and workshops with colleagues locally, nationally and internationally' [22].

Part of a quality management system is to have clear defined roles among staff; functions and responsibilities should be defined. The responsibilities should be allocated and understood by all involved staff. One must also make sure that the staff have the experience and competencies required by the different roles. A continuous competence development is of high importance to be able to keep up with all new modalities, technologies, etc. being introduced into the radiotherapy community.

One may also add that training and education must include information regarding the risks involved in the high-tech environment that radiotherapy is and on what has happened historically and what actions have been taken to improve safety.

A part that is often over-seen in many reports from adverse advents is the management of the involved staff. In a no-blame culture it is mandatory to have an organisation backing up the individual as well as the whole group of co-workers. Focus must be on the organisational and systemic conditions that actually lead to the adverse event and not the individual's mistake.

'Given the right set of circumstances, any of us can make a mistake' (from the Canadian report analysing the Ottawa accident [9]). However, having a quality management system implemented in radiation oncology (in principle health care overall) set-up with a defence-in-depth will reduce the number of incidents to pass and affect the patient. Defence-in-depth shall ensure that if one barrier, quality control is failing that subsequent independent barriers are available [23].

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